HDB Monthly Rent Prediction

Progress Report

CS5228 Competition Project

Exploratory analysis and first validation evidence

150,000 labeled records | 50,000 competition inputs

Observed approval dates: 2021-01 to 2026-07

Local analysis of the supplied project data

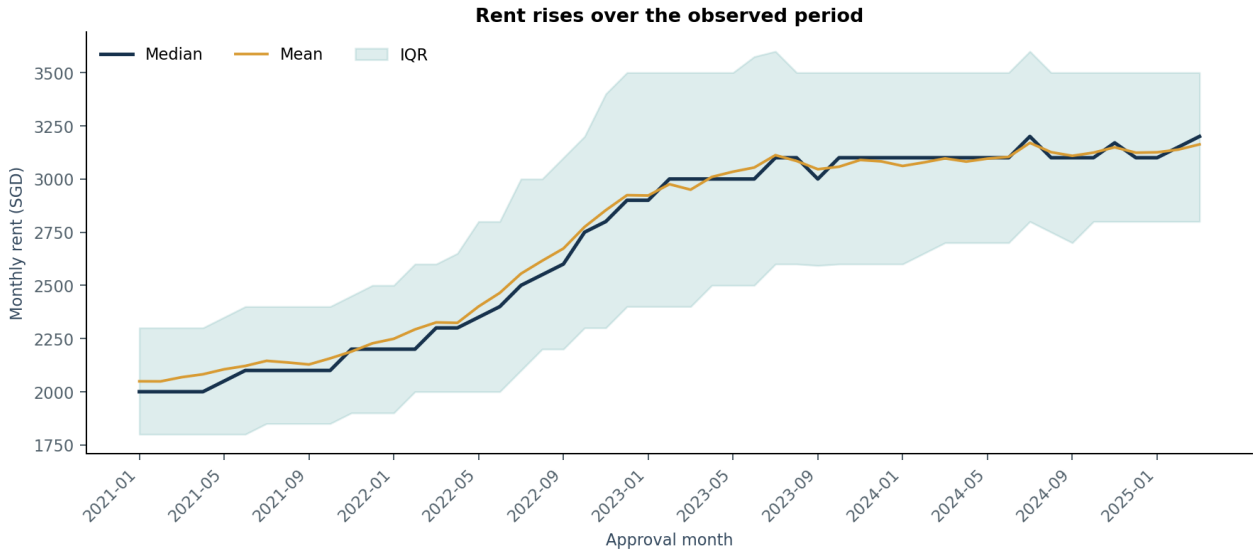
Project question and analysis plan

Question

How well can property, location and approval-time information predict monthly HDB rent for future records?

Current design

- Audit the supplied data and label distributions.
- Compare random and chronological holdouts.
- Test property and location features before adding date-aware economic series.



Source: data/train.csv, data/test.csv; EDA protocol in scripts/eda_analysis.py

Competition data and prediction boundary

Labeled training	**150,000 rows**	2021-01 to 2025-03
Competition test	**50,000 rows**	2025-03 to 2026-07
Target	**MONTHLY_RENT**	SGD per month
Inputs	**10 shared fields**	home, place, approval month

The test window extends beyond the labeled period, so future-period validation is central to estimating competition performance.

Source: data/train.csv and data/test.csv; row counts and dates: reports/eda/summary.json

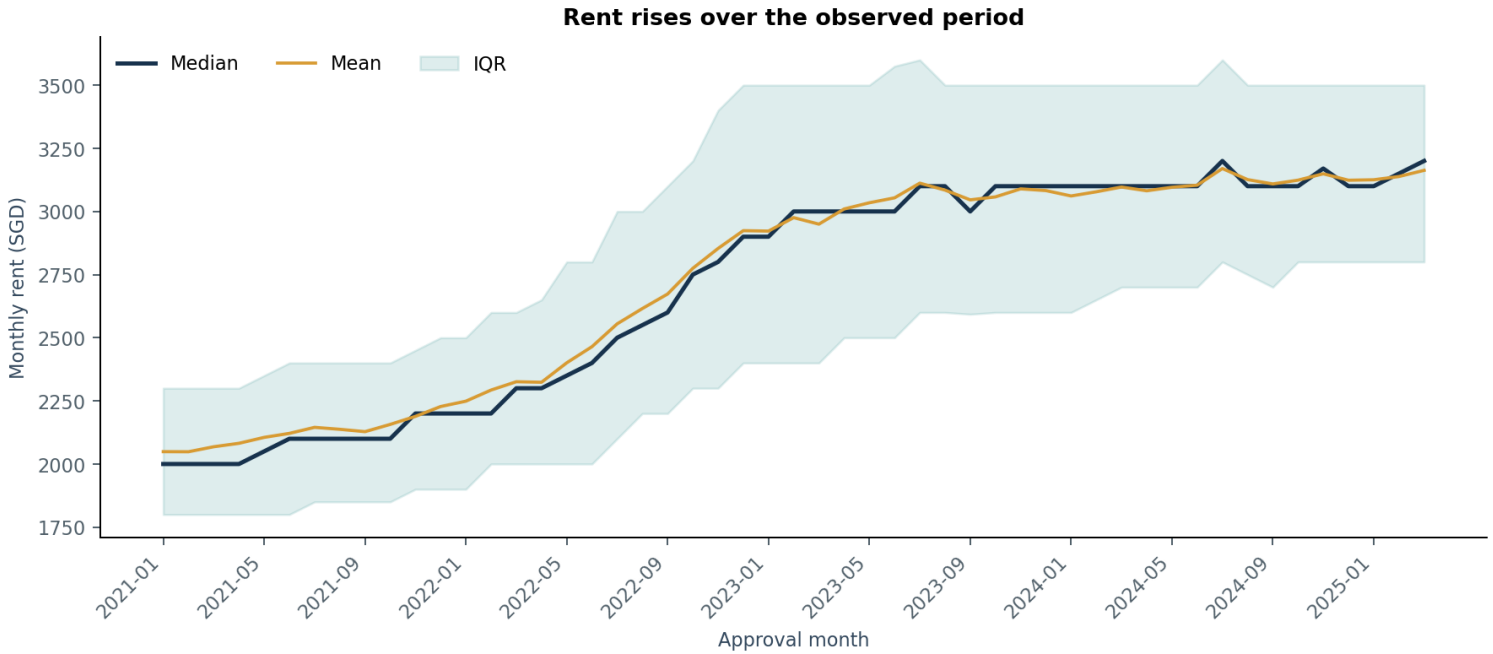
Target distribution



- Mean $2,719; median $2,700 per month.
- Middle 50%: $2,100 to $3,200.
- Skewness 0.39; observed range $300 to $7,600.
- Mild right skew does not by itself justify a log target; compare validation errors before choosing a transform.

Source: reports/eda/summary.json; reports/eda/figures/01_target_distribution.png

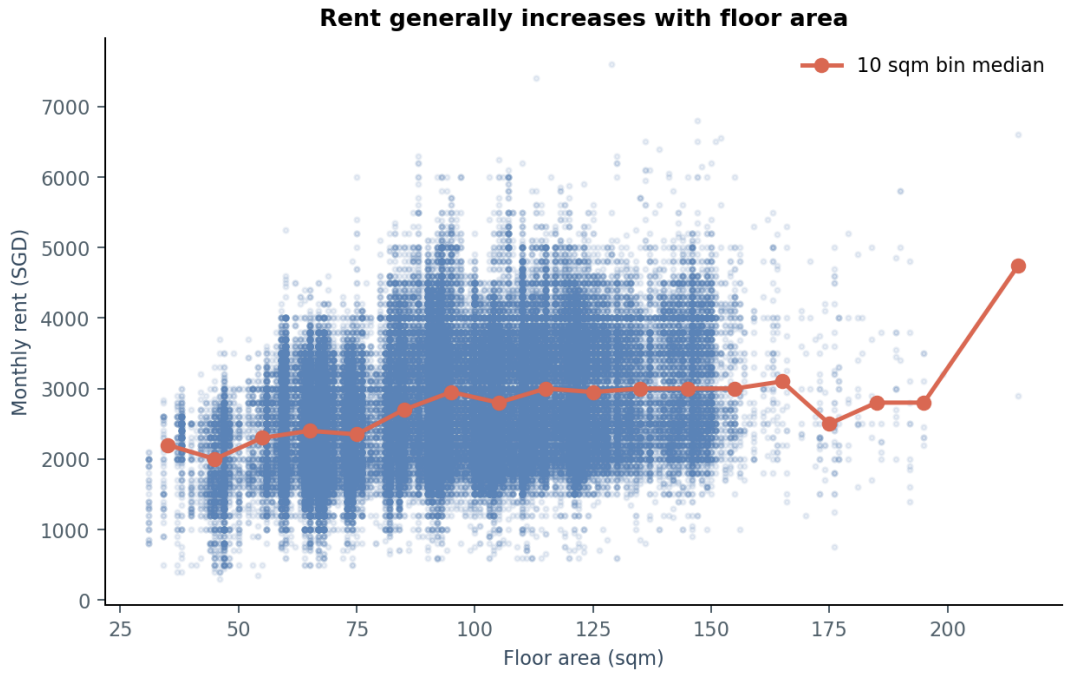
Rent levels change across approval time



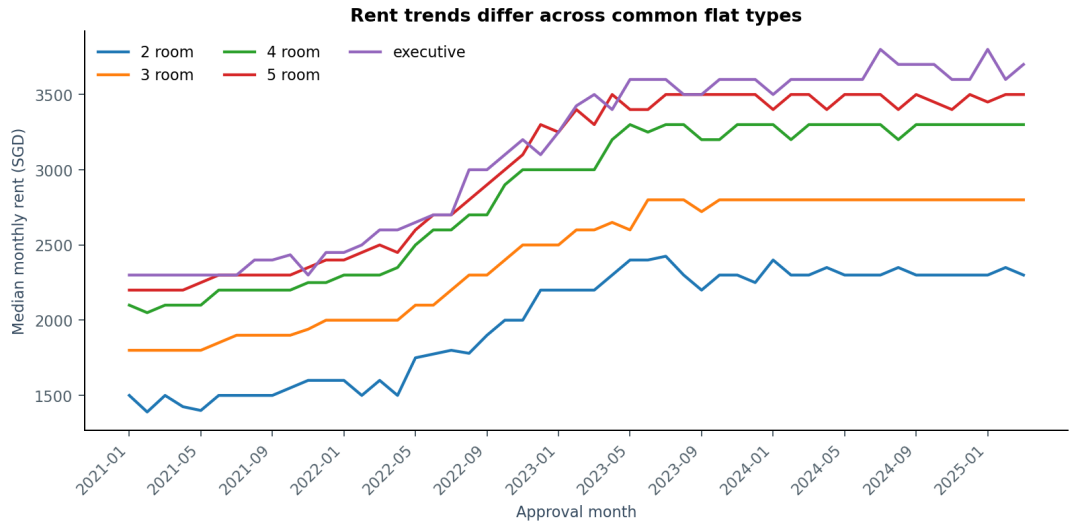
- Annual median: $2,100 in 2021 to $3,150 in 2025.
- The 2025 figure covers 8,924 rows through March only.
- Time is a first-order feature candidate and a reason to report chronological validation.

Source: reports/eda/tables/annual_rent_trend.csv; monthly series in reports/eda/tables/monthly_rent_trend.csv

Property and flat-type patterns



Area spans 31-215 sqm (median 92); rank association with rent is 0.32. Large-area bins are sparse.



Flat-type rent trends rise over time, with different levels by type. Normalize spelling before encoding categories.

Source: reports/eda/figures/05_rent_vs_floor_area.png and 09_monthly_rent_by_flat_type.png

Block reference data provides full coordinate coverage

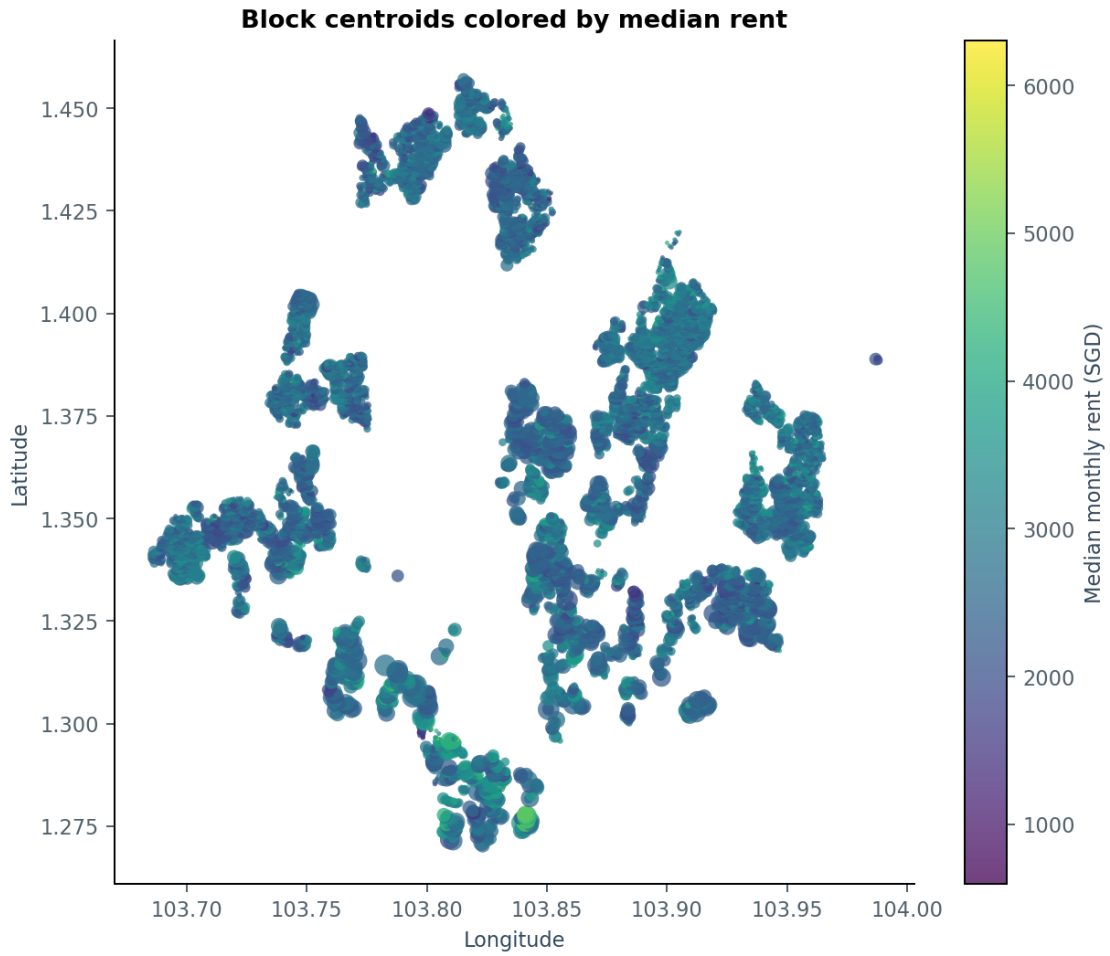
100%

train and test rows matched
a HDB block key

9,464

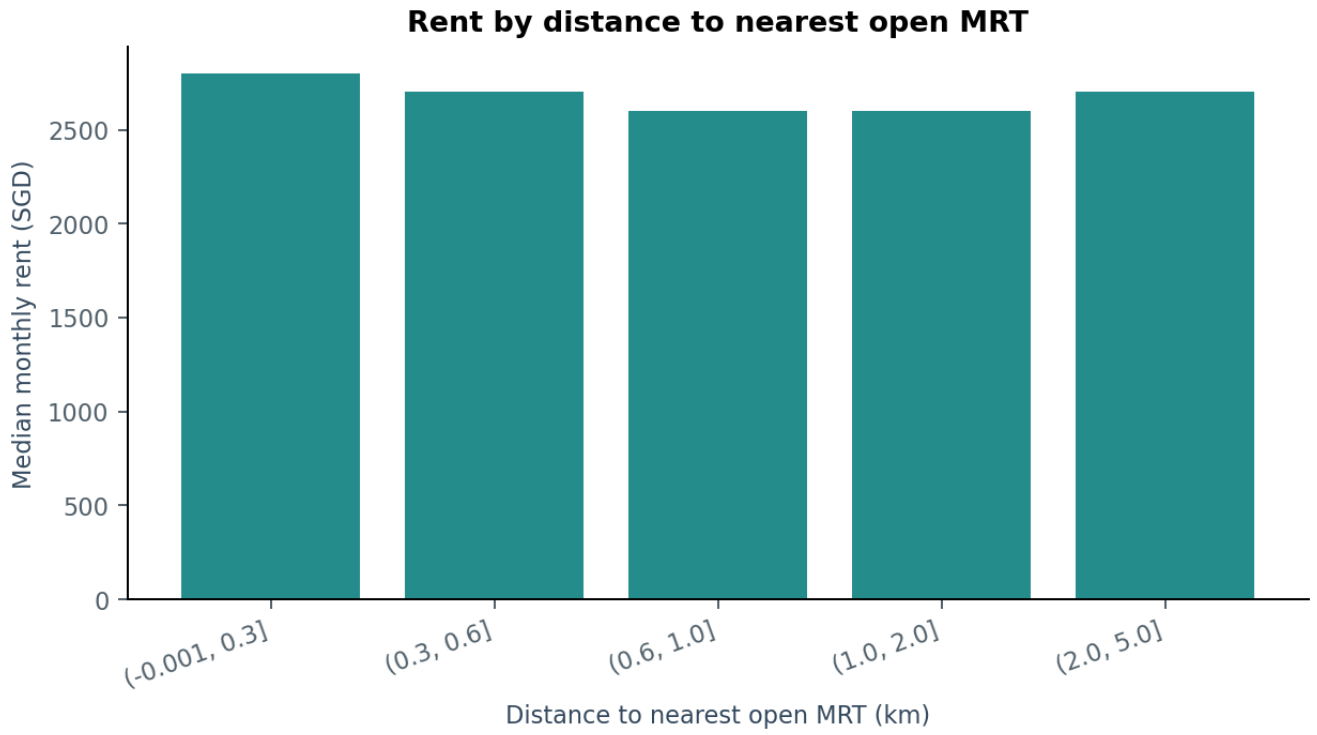
unique block keys

The map shows median rent at block centroids. Use location as a feature group and test its incremental value.



Source: data/auxiliary/sg-hdb-block.csv; map and join diagnostics: reports/eda/summary.json

Raw distance associations are modest



- Nearest open MRT raw Spearman with rent: -0.086.
- Rent medians differ across distance bands ($2,800 near MRT; $2,600 at 0.6-1.0 km).
- Town, flat type, area and time are mixed in these unadjusted comparisons; do not read them as an MRT premium.

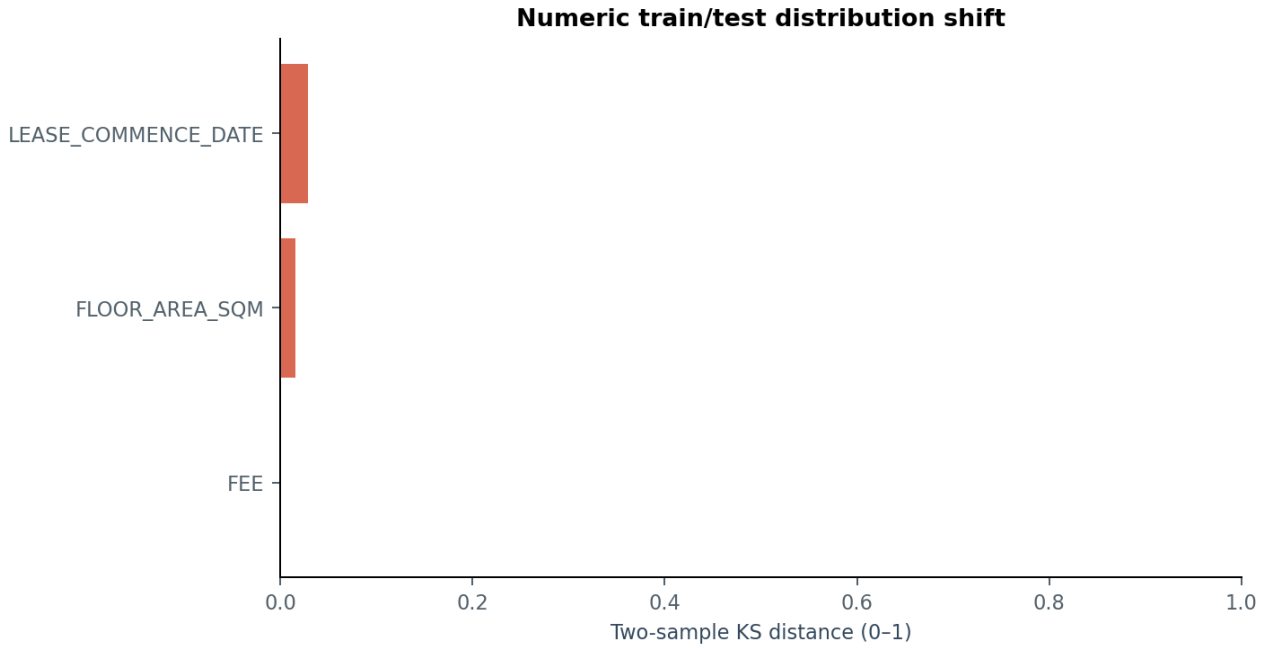
Source: reports/eda/figures/10_rent_by_open_mrt_distance.png; reports/eda/tables/location_feature_summary.csv

Data quality decisions preserve prediction-row integrity

No missing values	in the supplied train and test columns
Constant fields	FURNISHED = yes; FEE = 0
Label cleanup	FLAT_TYPE 11 to 6 categories; STREET 1146 to 576
Repeated records	425 exact duplicate train rows; 5,508 exact duplicate test rows. Of 13,434 repeated feature groups, 13,191 have different rents.
Policy	Keep every test row in original order; do not deduplicate submission inputs

Source: reports/eda/tables/data_quality.csv and categorical_normalization.csv

The competition test set is a future-time test



- Train: 2021-01 to 2025-03.
- Test: 2025-03 to 2026-07.
- Most test dates fall after the final labeled month; no target labels exist to measure 2026 error.
- Sixteen test months are unseen in training (49,620 rows). Numeric KS distances are small for floor area and lease commencement year.

Source: reports/eda/summary.json; reports/eda/tables/train_test_drift.csv

Auxiliary data creates useful features with timing constraints

HDB blocks	**9464**	coordinates, completion year, planning area
MRT stations	**243**	open and planned station points
Malls / schools	**113 / 337**	nearest-distance features
COE / stocks	**539 / 91,642**	monthly or daily time series

Macro series are not merged in this EDA. Any later use must aggregate only information available by each approval date; same-month values can be published after the rental record. MRT status has no historical opening dates, so planned-station distances are current snapshots rather than time-valid features.

Source: data/auxiliary/*.csv; audit: reports/eda/tables/auxiliary_data_audit.csv

Recent history improves the temporal baseline

Validation	Fit window	Rows	MAE	RMSE
Random 20%	all dates	30,000	$567	$668
Jan 24-Mar 25	all history	44,163	$787	$888
Jan 24-Mar 25	previous 12m	44,163	$384	$518
Jan-Mar 25	previous 12m	8,924	$345	$491

On the same future window, restricting group medians to the previous 12 months reduces RMSE by 41.6% versus using all history. The random holdout mixes years and is not directly comparable. Scores are local MAE/RMSE; the official competition metric was not verified.

Source: reports/eda/tables/baseline_validation.csv; median predictor by TOWN x FLAT_TYPE

Current decisions and open challenges

- Use chronological holdouts as the primary model-selection evidence; keep random split as a diagnostic comparison.
- Normalize text labels and preserve all test rows in source order.
- Start with property, normalized categories and approval time; add spatial features as a separate ablation.
- Treat nearest-place correlations as descriptive; compare adjusted model performance before attributing value.
- Remaining challenge: the labeled data stop in March 2025, while most competition inputs are later.

Evidence: reports/eda/summary.json and reports/eda/tables/*.csv

Next steps | build and compare models

- Extend rolling-origin validation to see whether recent-window baseline gains persist across cutoffs.
- Compare a simple linear model with a tree ensemble on the same split and feature set.
- Add feature groups in sequence: property, time, then spatial; record ablation results.
- Check residuals by month, town and flat type to identify systematic errors.

Planned work; validation windows will be fixed before model comparison

Next steps | finalize evaluation and report

- Test date-aware COE or stock features only if availability timing can be established.
- Select and tune the strongest validated approach; keep the experiment log and preprocessing reproducible.
- Generate competition predictions without changing test row count or order; validate the submission schema.
- Write up the final method, measured results, interpretation, limitations and remaining uncertainty.